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Mobile and wireless networks

Research paper analysis - Physical Layer Secret Key Generation

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1 Introduction

The growth of distributed wireless networks, including 5G and the Internet of Things (IoT), has escalated the risk of malicious attacks against message confidentiality. While symmetric-key encryption is preferred for resource constrained devices due to its low complexity, it requires an initial distribution of secret keys. Traditional physical layer security exploits natural channel reciprocity and randomness, yet these methods often yield ultra-low key rates in static environments where mobility is limited. This leaves indoor IoT networks particularly vulnerable to exploitation.

In their 2020 paper, *Physical Layer Secret Key Generation in Static Environments*, Nasser Aldaghri and Hessam Mahdavifar proposed a low-complexity solution called induced randomness to overcome these limitations. Two devices independently generate local random symbols, using the uniqueness of the wireless channel to facilitate high-rate key generation even when the environment is immobile. This approach is analyzed across two scenarii: direct communication and communication through an untrusted relay. Through a multi-stage protocol involving quantization, reconciliation via secure sketch, and privacy amplification, the authors provide an analytical evaluation of its robustness against passive eavesdroppers.

This report will review the steps of the proposed protocol, with an emphasis on building intuition for the underlying mathematical principles.

2

System model

This section introduces system models for the 2 types of secret key generation (SKG) protocol and its later security/performance analysis. We consider two architectures: (i) direct Alice–Bob communication and (ii) communication via an amplify-and-forward (AF) relay Carol. A passive eavesdropper Eve can overhear all wireless transmissions and the public discussion.

2.1 Entities, channels, and general assumptions

Legitimate parties

Alice and Bob are the two legitimate endpoints. They communicate over authenticated wireless links. It means that Alice and Bob can be trusted but the wireless communication is not trusted: an external listener may overhear transmissions.

Eavesdropper

Eve is assumed to be *passive*: she listens to all wireless transmissions between Alice and Bob, but she does not modify packets.

Public discussion channel

In addition to the wireless channel, Alice and Bob are assumed to have a noiseless public channel that Eve can access. This channel is used in later protocol steps (reconciliation and consistency checking) and can be realized with robust modulation/coding schemes.

Fading and reciprocity

Wireless channels are modeled as fading channels with complex coefficients. One of the most important characteristic of SKG is *reciprocity*: if Alice and Bob exchange signals within the channel coherence time and with adequate synchronization, the channel observed in the two directions is approximately the same.

OFDM vector model

The paper assumes OFDM to increase the key rate. For one OFDM symbol, signals can be represented as vectors across N subcarriers. In this representation, per-subcarrier fading becomes an element-wise multiplication (parallel narrowband channels).

2.2 Direct Secret Key Generation

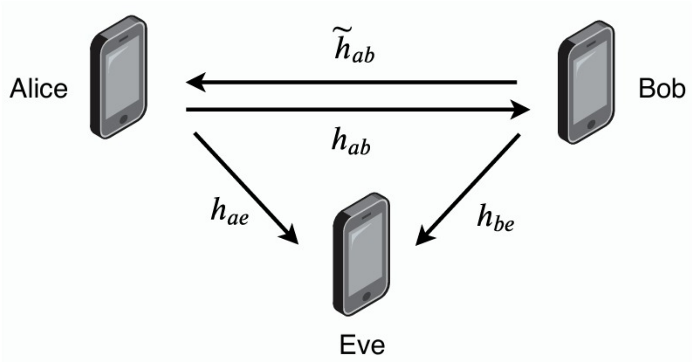

Figure 2.1: System model for direct secret key generation.

Assume OFDM with N subcarriers so that each subcarrier experiences a narrowband (flat) fading coefficient. Over one OFDM symbol time t, the received vectors can be written as

$$\mathbf{y}_{\text{Alice}}(t) = \mathbf{x}_{\text{Bob}}(t) \circ \tilde{\mathbf{h}}_{ab}(t) + \mathbf{n}_a(t), \tag{2.1}$$

$$\mathbf{y}_{\text{Bob}}(t) = \mathbf{x}_{\text{Alice}}(t) \circ \mathbf{h}_{ab}(t) + \mathbf{n}_b(t). \tag{2.2}$$

where $\circ$ denotes the element-wise product. Channel reciprocity within the coherence time implies $\mathbf{h}_{AB}(t) \approx \tilde{\mathbf{h}}_{AB}(t)$, which is the physical property enabling Alice and Bob to obtain highly correlated observations.

In addition, Alice and Bob exchange protocol messages over a noiseless public channel that Eve can listen to.

2.3 Secret Key Generation Using a Relay

Alice first transmits $\mathbf{x}_{\text{Alice}}(t)$ to Carol. Carol is assumed compliant (it forwards without tampering) but untrusted (can inspect the content).Carol receives

$$\mathbf{y}_{\text{Carol}}(t) = \mathbf{x}_{\text{Alice}}(t) \circ \mathbf{h}(t) + \mathbf{n}_r(t), \tag{2.3}$$

where $\mathbf{h}(t)$ is the Alice–Carol channel vector and $\mathbf{n}_r(t)$ is receiver noise at the relay. Carol then amplifies the signal with an amplification vector $\boldsymbol{\alpha}$ and forwards it to Bob. Bob receives

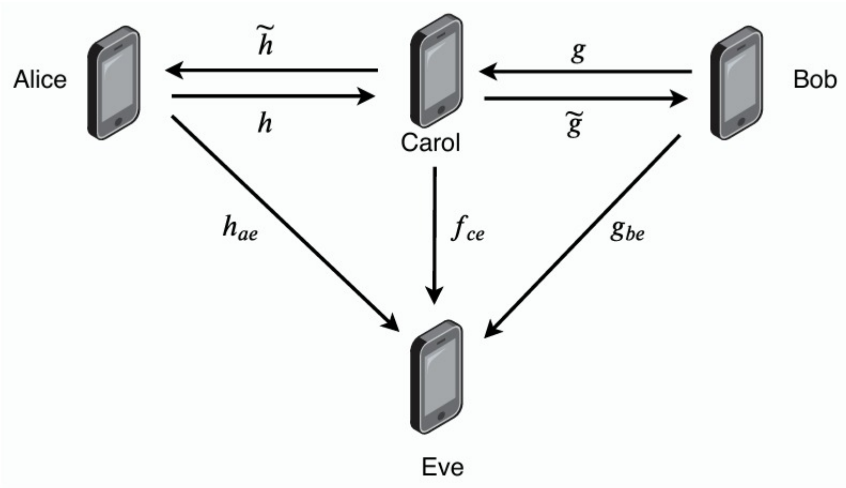

Figure 2.2: System model for relay-based secret key generation

$$\mathbf{y}_{\text{Bob}}(t) = \boldsymbol{\alpha} \circ \mathbf{y}_{\text{Carol}}(t) \circ \tilde{\mathbf{g}}(t) + \mathbf{n}_b(t) \tag{2.4}$$

$$= \boldsymbol{\alpha} \circ \big(\mathbf{x}_{\text{Alice}}(t) \circ \mathbf{h}(t) + \mathbf{n}_r(t)\big) \circ \tilde{\mathbf{g}}(t) + \mathbf{n}_b(t), \tag{2.5}$$

where $\tilde{\mathbf{g}}(t)$ is the Carol–Bob channel vector. Similarly, when Bob transmits to Alice through Carol, Alice receives

$$\mathbf{y}_{\text{Alice}}(t) = \boldsymbol{\alpha} \circ \big(\mathbf{x}_{\text{Bob}}(t) \circ \mathbf{g}(t) + \mathbf{n}_r(t)\big) \circ \tilde{\mathbf{h}}(t) + \mathbf{n}_a(t), \tag{2.6}$$

with $\mathbf{g}(t)$ the Bob–Carol channel and $\tilde{\mathbf{h}}(t)$ the Carol–Alice channel.

2.4 Evaluation metrics

- **Bit Generation Rate (BGR)**: number of quantized bits generated per packet.
- **Bit Mismatch Rate (BMR)**: fraction of mismatched bits between Alice's and Bob's quantized sequences (and similarly at Eve).

3

Proposed protocols

The paper proposes a secret key generation (SKG) protocol suitable for static or slowly varying channels, called Induced Randomness. Unlike traditional physical-layer security methods that rely solely on naturally occurring channel randomness, this protocol injects local randomness at each terminal to guarantee high-entropy key generation, even when the channel itself varies little, or in other words contains sparse scattering. The protocol operates in four distinct stages : **Induced randomness exchange**, **Quantization**, **Information reconciliation** and **Privacy amplification / consistency checking**. The first two stages contribute to producing identical, secure keys at Alice and Bob while the last ones ensure leakage limitation to an eavesdropper.

3.1 Induced Randomness Exchange

As briefly mentioned, in static environment, wireless channels are subject to very slow temporal variation which limits the required randomness rate to generate the secured key. To address this fundamental issue, the protocol proposes to share random sequences generated independently at each endpoint through the unique channel observed by Alice and Bob. The benefit of this random number communication relies on the highly correlated observation of the transmitted and received signals at endpoints, which appear as stochastic variables to a passive attacker. The randomness exchange differs depending on the chosen scenario and will be detailed in the following sub-sections.

3.1.1 Direct Communication Scenario

Alice and Bob generate independent random vectors $\mathbf{s}_i$ and $\mathbf{v}_i$ from an M-QAM constellation. They exchange these vectors over a direct wireless channel. Ignoring the noise for simplicity, the received signals are:

$$\mathbf{y}_a \approx \mathbf{v}_i \circ \tilde{\mathbf{h}}_{ab} \tag{3.1}$$

$$\mathbf{y}_b \approx \mathbf{s}_i \circ \mathbf{h}_{ab} \tag{3.2}$$

where $\mathbf{h}_{ab} \approx \tilde{\mathbf{h}}_{ab}$ denotes reciprocal channel coefficients vector, and $\circ$ the element-wise product. Each party then multiplies its transmitted vector with its received vector to produce correlated sequences:

$$\mathbf{w}_{i,ab} = \mathbf{s}_i \circ \mathbf{y}_a \approx \mathbf{s}_i \circ \mathbf{v}_i \circ \tilde{\mathbf{h}}_{ab} \tag{3.3}$$

$$\tilde{\mathbf{w}}_{i,ab} = \mathbf{v}_i \circ \mathbf{y}_b \approx \mathbf{v}_i \circ \mathbf{s}_i \circ \mathbf{h}_{ab} \tag{3.4}$$

Since $\mathbf{h}_{ab} \approx \tilde{\mathbf{h}}_{ab}$ due to reciprocity, $\mathbf{w}_{i,ab} \approx \tilde{\mathbf{w}}_{i,ab}$. These complex sequences, highly correlated, form the basis for the secret key generation.

3.1.2 Relay-Based Scenario

Without a direct link, the communication relies on an intermediate, an untrusted relay which amplifies and forwards signals without tampering them. Alice and Bob transmit random vectors $\mathbf{s}_i$ and $\mathbf{v}_i$ simultaneously. The relay receives the random signals, amplifies them by a factor α (chosen to secure wanted SNR) and broadcasts the sum.

$$\mathbf{y}_{i,a} = \alpha\big(\mathbf{s}_i \circ \mathbf{h}_i + \mathbf{v}_i \circ \mathbf{g}_i + n_{i,1,r}\big) \circ \tilde{\mathbf{h}}_i + n_{i,2,a}, \tag{3.5}$$

$$\mathbf{y}_{i,b} = \alpha\big(\mathbf{s}_i \circ \mathbf{h}_i + \mathbf{v}_i \circ \mathbf{g}_i + n_{i,1,r}\big) \circ \tilde{\mathbf{g}}_i + n_{i,2,b}, \tag{3.6}$$

where $\tilde{\mathbf{h}}$, $\mathbf{h}$, $\mathbf{g}$ and $\tilde{\mathbf{g}}$, denote, respectively, Alice-Carol's channel, Carol-Alice's channel, Bob-Carol's channel and Carol-Bob's channel.

Using known transmitted signals and estimated channels to the relay, Alice and Bob cancel self-interference to extract a highly correlated term, preventing signal reconstruction by the untrusted relay - Equation 3.8.

$$\mathbf{w}_{i,ab} = \mathbf{s}_i \circ \mathbf{v}_i \circ \mathbf{g}_i \circ \tilde{\mathbf{h}}_i + \bar{n}_{i,2,a}, \tag{3.7}$$

$$\tilde{\mathbf{w}}_{i,ab} = \mathbf{s}_i \circ \mathbf{v}_i \circ \tilde{\mathbf{g}}_i \circ \mathbf{h}_i + \bar{n}_{i,2,b}, \tag{3.8}$$

where $\mathbf{w}_{i,ab}$ denotes Alice's reconstructed signal and $\tilde{\mathbf{w}}_{i,ab}$, Bob's reconstructed signal.

3.2 Quantization

The correlated complex sequences $\mathbf{w}_{i,ab}$ and $\tilde{\mathbf{w}}_{i,ab}$ are converted to binary sequences via sorting-based quantization for key extraction:

1. Sort complex measurements as real and imaginary intervals (in total, 2**xN**carrier).

2. Compute the range [min, max].

3. Divide range into 2^{δ} uniform intervals (δ = quantization resolution).

4. Map each interval to a Gray-code bit sequence.

The resulting bit sequences for Alice and Bob are denoted $\mathbf{q}_{i,a}$ and $\mathbf{q}_{i,b}$, respectively.

3.3 Information Reconciliation

To correct noise-induced mismatches, non-ideal channel reciprocity and quantization errors between $\mathbf{q}_{i,a}$ and $\mathbf{q}_{i,b}$, the protocol makes use of a secure sketch algorithm relying on a convolutional encoder. This secure sketch ensures to correct bits mismatches between Alice and Bob by minimizing the information leakage generated through the correction codeword communication. First, Alice generates an independent random bits sequence $\mathbf{r}$ of dimension $\mathbf{k} \leq length(\mathbf{q}_{i,a})$, encodes it using a convolutional encoder (combination of register shift and addition modulo 2), producing the codeword $Enc(\mathbf{r})$ of same length as $\mathbf{q}_{i,a}$. Alice, then, computes the secure sketch.

$$SS = \mathbf{q}_{i,a} \oplus Enc(\mathbf{r}) \tag{3.9}$$

Which is transmitted over a noiseless public channel, accessible to Eve too.
Once the signal is received, Bob combines the secure sketch SS with his own sequence, $\mathbf{q}_{i,b}$.

$$SS \oplus \mathbf{q}_{i,b} = Enc(\mathbf{r}) \oplus (\mathbf{q}_{i,a} \oplus \mathbf{q}_{i,b}) \tag{3.10}$$

The strength behind SS communication to Bob lies in the computation of the encoded sequence $Enc(\tilde{\mathbf{r}}) = Enc(\mathbf{r}) \oplus \mathbf{e}$, corrupted by a small number of errors $(\mathbf{q}_{i,a} \oplus \mathbf{q}_{i,b})$. Conditioning that the error does not exceed the correction capability of the code (function of r redundancy, dimension), Bob can recover the true r, re-encode it, and reconstruct the true $\mathbf{q}_{i,a}$.

From the security point of view, the secure sketch usage does not come with a null cost. The sequence entropy seen by the Eve is reduced by a factor $(length(\mathbf{q}_{i,a}) - k)$. This entropy reduction, or, in other words, information leakage, will be removed through privacy amplification.

3.4 Privacy Amplification and Consistency Checking

The entropy reduction introduced during the reconciliation stage is mitigated through the usage of Universal Hash Functions (UHF) to ensure the stochasticity of key generation seen by Eve. The basic idea of the UFH is to map a sequence of bits $\mathbf{q}$ into a output space of less dimension $\mathbf{t}$ which will be used as the new key by ensuring that the probability for two inputs $\mathbf{u}$ and $\mathbf{y}$ to generate the same output via a random hash function is less than $\frac{1}{t}$. To guarantee that Alice and Bob generates the same hashed key, both will divide their reconciled sequences by **two**. The first half will be used to define the hashed function to use and the second half will be the input. Mathematically, it boils down to

$$\mathbf{K}_{ab} = h_{\mathbf{q}_{i,1}}(\mathbf{q}_{i,2}). \tag{3.11}$$

Where $\mathbf{K}_{ab}$ is the generated key through the hashed function of length $\frac{\mathbf{n}}{2}$.

The validation of Alice's and Bob's new key are done by re-hashing in a same manner the hashed key, transmit them in the public channel and confirm the sequences match. If the validation failed, the hashing session is repeated.

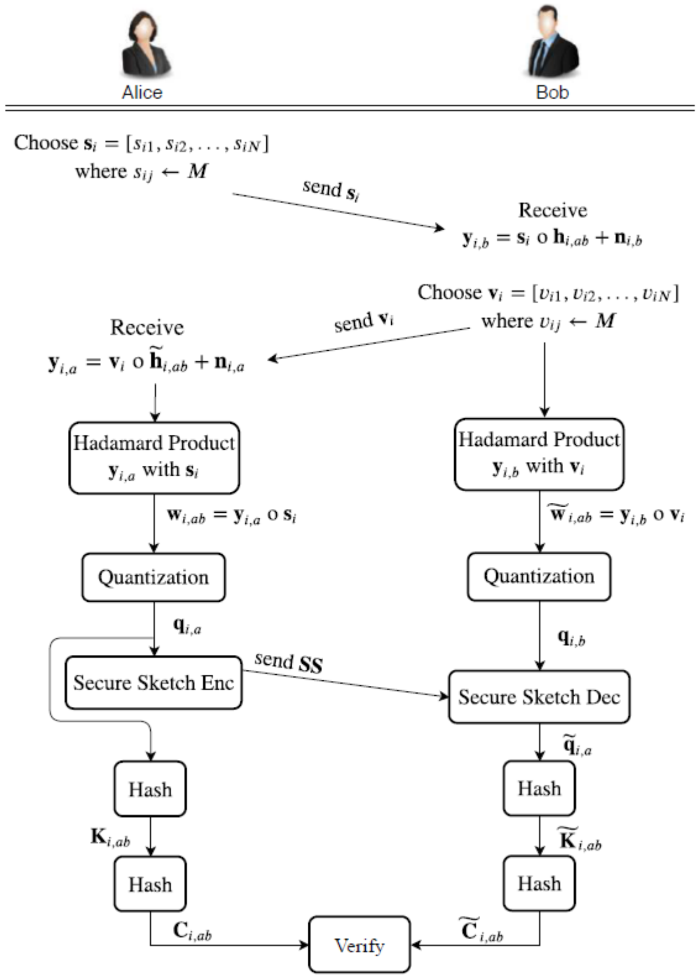

Figure 3.1: Overview of the Direct Secret Key Generation Protocol.

4 Attack model and protocol resilience

The section focuses on the resilience of the key generation protocol discussed during the previous chapters. Our attacker, Eve, is assumed to be a passive eavesdropper for both communication architectures. For each scenario, we will discuss the level of information Eve can extract from her observations and derive the upper bound on the probability to have a successful eavesdropper attack. While mathematical developments in this paragraph rely on information theory tools such as conditional entropy and mutual information, our objective is not to reproduce the paper detailed mathematical derivations but rather provides a intuitive interpretation of those. In particular, we will emphasize how the information leakage securities (quantization, channel decorrelation and the privacy amplification) take parts in the probability upper bound calculation.

4.1 Mutual information

We first need to look into the signals seen by Eve to grasp the potential danger. The equations 4.1 and 4.2 expand mathematically the signal as combination of element-wise multiplication.

$$w_{i,\text{ed},1} = s_i \circ v_i \circ h_{i,\text{ae}} + n_{i,\text{ae}}, \tag{4.1}$$

$$w_{i,\text{ed},2} = s_i \circ v_i \circ h_{i,\text{be}} + n_{i,\text{be}}. \tag{4.2}$$

Where $\mathbf{h}_{i,\text{ae}}$ and $\mathbf{h}_{i,\text{be}}$ denotes, respectively, Alice-Eve channel and Bob-Eve channel.

One can understand that the observation being successful will depend entirely on the correlation between either the channel $h_{i,\text{ae}}$ and $h_{i,\text{ab}}$ or between the channel $h_{i,\text{be}}$ and $h_{i,\text{ab}}$.

We discussed previously about the spatial correlation of the channel and the reciprocity properties between the channel observed by the sender and the receiver. It would be interesting to quantify the correlation between the channel seen by our attacker Eve and Bob-Alice|Alice-Bob channel as it intervenes in the probability computation of leaked information seen from Eve side.

From channel communication course, we know that the magnitude of a channel observed at distance d, can be built as a statistical model realisation (from Rice or Rayleigh pdf model). The spatial correlation between two channels realisation considering small scale fading can be expressed as a Bessel function. Mathematically it boils down to the equation 4.3.

$$J_0(d) \approx \begin{cases} 1, & d \ll \lambda, \\ 0, & d \approx 0.38\,\lambda, \\ \text{oscillatory decay}, & d > \lambda. \end{cases} \tag{4.3}$$

Where λ and **d** denote respectively the signal wavelength and the distance between the attacker and the closest endpoint (Alice or Bob).

Intuitively, this result shows that the wireless channels observed by two receivers separated by more than approximately half a wavelength (general rule of thumb) are weakly correlated. As a consequence, an eavesdropper located at such a distance experiences an essentially independent channel realization. This spatial decorrelation, confirming the channel randomness seen by a sufficient distant observer, is the fundamental property exploited for physical layer key generation.

We also know that channel coefficients are complex numbers. From the statistical models, channel can be therefore decomposed in an imaginary sub-channel and real sub-channel independently and identically distributed as $\mathcal{N}(0, \sigma^2)$. From 4.4, the real and imaginary parts of Alice (or Bob) and Eve are correlated with ρ.

$$\rho = [J_0(d)]^2 \tag{4.4}$$

From the channel correlation and the channel model distribution, we can build the covariance matrices of both channels, h_{ae} and h_{ab} and the covariance matrice from joint channel (h_{ae},h_{ab}). These covariance matrices being used to calculate the mutual information in equation 4.8.

$$\mathbf{C}_{\text{Bob}} = \begin{pmatrix} \frac{\sigma_{\text{b}}^2}{2} & 0 \\ 0 & \frac{\sigma_{\text{b}}^2}{2} \end{pmatrix} \tag{4.5}$$

$$\mathbf{C}_{\text{Eve}} = \begin{pmatrix} \frac{\sigma_{\text{e}}^2}{2} & 0 \\ 0 & \frac{\sigma_{\text{e}}^2}{2} \end{pmatrix} \tag{4.6}$$

$$\mathbf{C}_{\text{Joint}} = \begin{pmatrix} \frac{\sigma_{\text{b}}^2}{2} & 0 & \frac{\rho\sigma_{\text{b}}\sigma_{\text{e}}}{2} & 0 \\ 0 & \frac{\sigma_{\text{b}}^2}{2} & 0 & \frac{\rho\sigma_{\text{b}}\sigma_{\text{e}}}{2} \\ \frac{\rho\sigma_{\text{b}}\sigma_{\text{e}}}{2} & 0 & \frac{\sigma_{\text{e}}^2}{2} & 0 \\ 0 & \frac{\rho\sigma_{\text{b}}\sigma_{\text{e}}}{2} & 0 & \frac{\sigma_{\text{e}}^2}{2} \end{pmatrix} \tag{4.7}$$

The mutual information, denoted $I(h_{\text{ae}}, h_{\text{ab}})$, can be expressed through the conditional entropy development in equation 4.8

$$I(h_{\text{ae}}, h_{i,\text{ab}}) = H(h_{\text{bk}}) + H(h_{\text{ek}}) - H(h_{\text{bk}}, h_{\text{ek}}) \tag{4.8}$$

Where $H(h_{\text{bk}})$ and $H(h_{\text{ek}})$ denote channel entropy seen by Bob and channel entropy seen by Eve. The entropies exhibit the level of randomness or uncertainty inherent to the channels.
By expressing the entropies as multivariate gaussian distrubition, the mutual information results into the final expression 4.12.

$$\begin{aligned} I(h_{\text{ae}}, h_{i,\text{ab}}) &= H(h_{\text{bk}}) + H(h_{\text{ek}}) - H(h_{\text{bk}}, h_{\text{ek}}), && (4.9)\\ &= \frac{1}{2}\log\bigl(\det(2\pi e\mathbf{C}_{Bob})\bigr) + \frac{1}{2}\log\bigl(\det(2\pi e\mathbf{C}_{Eve})\bigr) - \frac{1}{2}\log\bigl(\det(2\pi e\mathbf{C}_{Joint})\bigr), && (4.10)\\ &= \log\bigl(\pi e\sigma_{\text{b}}^2\bigr) + \log\bigl(\pi e\sigma_{\text{e}}^2\bigr) - \log\bigl((\pi e\sigma_{\text{b}}\sigma_{\text{e}})^2(1-\rho^2)\bigr), && (4.11)\\ &= -\log(1-\rho^2) && (4.12) \end{aligned}$$

The final equality convey that the level of information leakage does depend exclusively of the spatial correlation between the attacker channel and the communication channel, and therefore independent of the absolute channel power or, in other words, independent of the signal to noise ratio.

4.2 Upper bound probability of successful attacks

The mutual information defined, we have all the tools to compute the upper bound probability of successful attacks. For that, we must compute each increase of probability generated by the implemented security processes, namely the quantization process, the channel spatial correlation and the privacy amplification process. In a first time, we will consider the direct communication architecture and afterwards compute the upper bound for the communication through untrusted relay.

4.2.1 Direct communication

The probability of a successful Eve's eavesdropping attack is upper bounded as stated by the equation 4.13.

$$Pr(success) < (2^{-2\delta} + \sqrt{2I(h_b, h_e)})^N + 2^{-N\delta}. \tag{4.13}$$

With :

1. $2^{-2\delta}$ term providing the probability of successful guess of the binary representation after quantization. The factor 2 coming from the imaginary and real channel coefficient decomposition.

2. $\sqrt{2I(h_b, h_e)}$ increasing the probability of recovering the right bits given Eve's observation from her own channel.

3. the exponent N being the OFDM subcarriers number, each carrier seeing a narrowband flat channel.

4. $2^{-N\delta}$ term raising the upper bound due to the additional guess of the key generation during the privacy amplification process.

It is worth mentioning that the privacy amplification increases the upper bound probability of successful attack but removes the information leaked during the code-secure sketch process.

4.2.2 Communication through untrusted relay

As a remainder, in this communication architecture, the relay, Carol, is sending back the received signal amplified by a public factor α. The attacker, Eve, has access to **two** signals shown in the equations 4.14 and 4.15.

$$w_{i,\mathrm{e},1} = s_i \circ h_{i,\mathrm{ae}} + v_i \circ g_{i,\mathrm{be}} + n_{i,\mathrm{e1}}, \tag{4.14}$$

$$e_{i,3} = \alpha\big(s_i \circ h_i + v_i \circ g_i + n_{i,2,\mathrm{r}}\big) \circ f_{i,\mathrm{ce}} + n_{i,\mathrm{e2}}. \tag{4.15}$$

Where $w_{i,\mathrm{e},1}$ denotes as the combined signal received by Eve and transmitted by Alice and Bob separately. $e_{i,3}$ denotes as the signal received from Carol. Finally, $f_{i,\mathrm{ce}}$ represents Carol-Eve channel.

From the OFDM preambles symbols, the channel $f_{i,\mathrm{ce}}$ can be estimated leading to the recovery of the signal $w_{i,\mathrm{e},2}$, in 4.16, equivalent to the signal seen by Carol.

$$w_{i,\mathrm{e},2} = s_i \circ h_i + v_i \circ g_i + n_{i,2,\mathrm{r}} \tag{4.16}$$

The spatial correlation property of the channel is, in this case, not providing the same security against the eavesdropper attack as the direct communication. The reason is Eve can virtually be considered as having access to the same level of information as the relay. This comes from the fact that the signal $w_{i,\mathrm{e},2}$ can be retrieved completely. The mutual information contains in $I(w_{\mathrm{i},ab}; w_{i,\mathrm{e},1}, w_{i,\mathrm{e},2})$ is much more significant than in the *section 4.2.1.* As such, the upper bound cannot be express from the equation 4.13 as saturating to 1 and thus not giving relevant information about the level of security for the relay architecture. Therefore, the idea is to pass by the Fano's inequality developed in the equation 4.18.

$$\Pr(success) \leq \big(1 - \frac{H(q_a \mid w_{i,\text{e},1}, w_{i,\text{e},2}) - 1}{\log_2 |\mathcal{Q}_A|}\big)^N, \tag{4.17}$$

$$\Pr(success) \leq \big(1 - \frac{H(q_a) - I(w_{\text{i},ab}; w_{i,\text{e},1}, w_{i,\text{e},2}) - 1}{\log_2 |\mathcal{Q}_A|}\big)^N. \tag{4.18}$$

Where $H(q_a)$ and $\mathcal{Q}_A$ denote respectively as the quantized bits sequence entropy and the signal modulation (M-QAM in our case).

If the attacker, Eve, cannot deduce the key during the communication session, the remaining probability to rightly guess the hashed key is equal to $2^{-N\delta}$. The total upper bound probability is therefore stated in the equation 4.19.

$$\Pr(C_N) \leq \big(1 - \frac{H(q_a) - I(w_{\text{i},ab}; w_{i,\text{e},1}, w_{i,\text{e},2}) - 1}{\log_2 |\mathcal{Q}_A|}\big)^N + 2^{-N\delta}. \tag{4.19}$$

4.2.3 Numerical attack probabilities

Let's consider communication with the following parameters.

1. The quantization parameter $\delta = 2$,
2. 16 OFDM subcarriers,
3. 64-QAM modulation,
4. Eve located away from Alice, Bob and Carol by at least $\frac{\lambda}{2}$.

The upper bound probability of successful passive eavesdropping is equal to 2^{-31} and $2^{-10.57}$, respectively for direct communication and through relay. In comparison with the protocol proposed in *Manipulatable wireless key establishment* (2017) for static environments, maximum probability of key acquisition by Eve is in ranges from 0.09% to 0.47%.

5 Results and discussion

The results section focuses on the bit generation rate (BGR) and the bit mismatch rate (BMR), as these metrics enable a direct comparison between the proposed algorithm and existing secret key generation protocols but also state the protocol limitations in real conditions. We will conclude by a critical analysis of the protocol and its limitations.

The research paper compares the results obtained by three different scenarii. For all the sceanrii, the channel used OFDM parallel transmission with 16 subcarriers and QAM symbols, the quantization after transmission is fixed at $\delta = 2$. The first scenario is a direct reciprocal channel communication. The second is a non light of side channel with realistic 5G mmWave channels coefficients (from the NYUSIM network simulator) in urban micro-cellular environment operating in 28 GHz with very low outdoor-indoor loss and Eve distant of 10 meters from Bob and Alice. The last one is relay-based architecture with only direct communication with the relay.

5.1 Bits generation rate

For all the scenarii, in the current modulation, quantization configuration and bandwidth division, the bit generation rate adds up to 64 bits/OFDM symbol. The paper mentions **64 bits/packet** expressing that the transmitted QAM symbols are average over all the OFDM symbols ensuring a minimization of the noise, optimizing the BMR per communication session. The key is generated after 4 sessions, current transmitted bits being added modulo 2 with the previous one to enhance secrecy (relevant for indoor NLOS communication - scenario 2). Regarding the performance, the paper protocol achieves comparable level of bits generation rate as protocols developed for dynamic environment. For static environment, previous protocols could generate between 0.5 bit per packet to 8 bits per packet.

For completeness, we suppose we want to generate a 128 bits secret key for the current 5G wireless network. For communication operating at 28GHz, we find 14 OFDM symbols per packet. Assuming a subcarrier spacing of 120 KHz, a 128 bits secret key can be established in few milliseconds, as shown in the equation 5.5. Thus, the key generation timing is equivalent to computation of key encryption methods such as RSA or handshaking methods and well within the timing constraint for 5G New Radio communication.

$$T_{\text{sym}} = \frac{1}{\Delta f} = \frac{1}{120\,\text{kHz}} \approx 8.33\ \mu\text{s}, \tag{5.1}$$

$$T_{\text{sym}}^{(\text{CP})} \approx 9\ \mu\text{s} \qquad\qquad \textbf{with cycle prefix}, \tag{5.2}$$

$$T_{\text{slot}} = 14 \times T_{\text{sym}}^{(\text{CP})} \approx 14 \times 9\ \mu\text{s} = 126\ \mu\text{s}, \tag{5.3}$$

$$T_{\text{32-bits}} = 4 \times T_{\text{slot}} \approx 4 \times 126\ \mu\text{s} \approx 0.5\ \text{ms}, \tag{5.4}$$

$$T_{\text{128-bits}} = 4 \times T_{\text{32-bit}} \approx 2\ \text{ms}. \tag{5.5}$$

5.2 Bits mismatch rate

The figure 5.1 illustrates the results for bits mismatch rate between Alice and Eve and compares it with the BMR between Alice and Bob. The comparison spans all the three setups. As remainder, the bits mismatch rate is the ratio between the computed quantized bits after transmission and the corresponding true quantized bits. We can observe that the BMR between Alice and Bob decreases with the SNR, while the BMR between Alice and Eve exhibits a very slow reduction behaviour and converges to a *plateau*. Nevertheless, the second scenario BMR between Alice and Eve is significantly lower than the other setups, decreasing in the SNR range from 5 to 15 dB before also reaching a *plateau*.

This moderate BMR in the indoor NLOS communication is attributed to sparse, or clustered scattering of the signal. The spatial correlation model of the physical channel rests on the assumption of infinite amount of multipath components (MPC) or scatters resulting to Rayleigh or Rice fading statistics. In 5GNR communication, the scatters are characterized by limited diffusion and dominant reflections, resulting to a small number of dominant paths. Under these sparse scattering conditions, the probability that Eve observes the same cluster of scatters as Alice and Bob is significantly higher than in the idealized case of infinite MPCs environment, and thus explaining the low BMR for this scenario.

Nevertheless, even for the NLOS communication, the BER between Alice and Eve converges to 50% after privacy amplification, thereby confirming the security robustness of the proposed protocol - Figure 5.2.

5.3 Limitations

The weakness of the proposed protocol relies on key assumptions resulting to potential limitations in case of protocol deployement.

1. **Passive eavesdropper attack :** The protocol assumes a passive eavesdropper attacker, not interfering with the communication between Alice and Bob. In case of active attacks from Eve, such as jam-

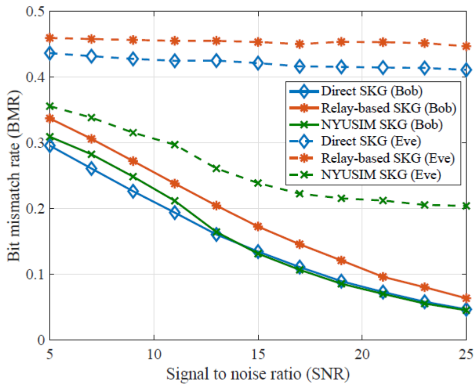

Figure 5.1: Bits mismatch rate - all setups comparison.

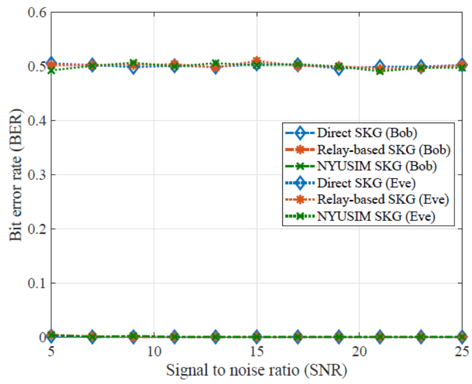

Figure 5.2: Bits mismatch rate - all setups comparison.

ming, the correlation is degraded, preventing Alice and Bob from reliable recovery of the introduced randomness from both parties.

2. **Channel stability :** The key generation process has been proved to be robust in static environment and remains valid for slow varying channels as long as the time coherence of the channel exceeds the key generation time. However, if the time needed to process the secret key is longer than the channel stability duration, the observations of Alice and Bob decorrelate, resulting to an increase of the BMR and potential key generation failure.

3. **Hardware synchronisation :** The protocol assumes a good synchronisation of Alice's and Bob's hardwares. Imperfect synchronisation may require concession on quantization resolution or on the rate of error-correcting code ensuring successful correlation and key derivation.

6 AI discussion

The usage of the AI in the framework of this report focuses on three pillars.

1. **Intuition behind the mathematical developments :** The major usage of the AI was centered around an assistant role for decrypting unknown mathematical simplification or decomposition. The mathematical foundations regarding the channel physics or information theory are well known for the electronics engineers. Nevertheless, some details like the Fano's inequality or the Bessel function of first order was just slightly brushed in our course without a deep interrogation about the detailed meaning. The AI in this case could give us detailed understanding and lead us to the intuition hidden behind the mathematics. It is worth to mention that while the AI provides mathematical guidance, we make sure to keep a critical thinking about the given explanation by confirming the alignement with what we have seen in course.

2. **Help about the report construction :** The second idea was to use the AI as support tool for the direction to give to this report. More specifically, it was to confirm the result or analysis-oriented report with a focus on the intuition behind the mathematical expressions instead of just summarize the research paper.

3. **Support for example construction and validation of results :** Finally, AI was used as a tool to construct simplified numerical examples providing support in the discussion of the scenarii results or for the critical analysis. For instance, the example about the key generation time was supported by the artificial intelligence to confirm the number of OFDM symbols contained in a 5GNR communication packet and double-checked in the literature afterwards.

7 Conclusion

The analyzed paper proposes a simple yet robust securized protocol for periodic secret key generation. The robustness has been evaluated by deriving the upper bound probability of successful passive attack, such as eavesdropping. Moreover, the protocol achieved similar performance of bits generation rate as protocols designed for dynamic environment communication, without introducing additional information leakage to the attacker. Finally, it was proved that the protocol is very well suited for indoor wireless communication, including 5GNR communication characterized by sparse scattering. This properties makes the protocol attractive for low resources devices, such as IoT devices, where conventional cryptographic methodology as RSA or TLS-based handshakes, are difficult to implement due to limited computational hardware and strict energy constraint of such devices.